

**Hyperbolic Functions**

Continuation....

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42. $\int \tanh^2 u \, du = u - \tanh u + C$

43. $\int \coth^2 u \, du = u - \coth u + C$

Trigonometry Substitution

44. $\int \sqrt{a^2 - u^2} \, du$ let: $u = a \sin \theta$

45. $\int \sqrt{a^2 + u^2} \, du$ let: $u = a \tan \theta$

46. $\int \sqrt{u^2 - a^2} \, du$ let: $u = a \sec \theta$

Integration By Parts

47. $\int u \, dv = uv - \int v \, du$

Wallis Formula

Formulated by John Wallis

48. $\int_0^{\frac{\pi}{2}} \sin^m \theta \cos^n \theta \, d\theta =$

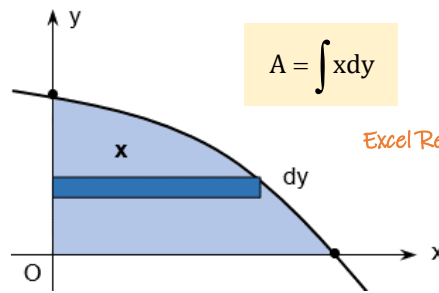
$$\frac{\left[\frac{(m-1)(m-3) \dots 1}{2} \right] \left[\frac{(n-1)(n-3) \dots 1}{2} \right]}{(m+n)(m+n-2)(m+n-4) \dots 1} \cdot \alpha$$

where:

$\alpha = \frac{\pi}{2}$ if both m and n are even

$\alpha = 1$ if otherwise

b. Using a horizontal differential strip



$$A = \int x \, dy$$

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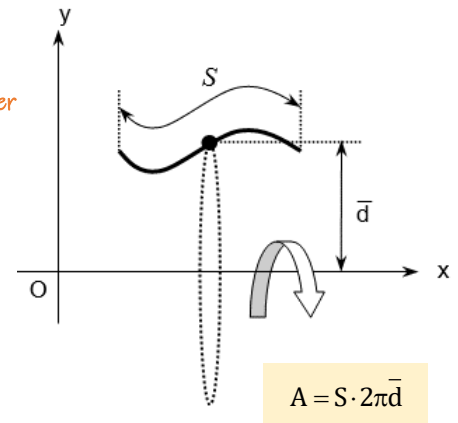
$$S = \int_{x_1}^{x_2} \sqrt{1 + \left(\frac{dy}{dx} \right)^2} \, dx$$

$$S = \int_{y_1}^{y_2} \sqrt{1 + \left(\frac{dx}{dy} \right)^2} \, dy$$

4. Surface Area

The surface area of a solid generated by revolving a curve about a certain axis may be calculated using the first proposition of Pappus. The **First Proposition of Pappus** states as follows:

"If an arc is rotated about an axis, it will generate a surface area equal to the product of the length of the arc and the circumference described its centroid."



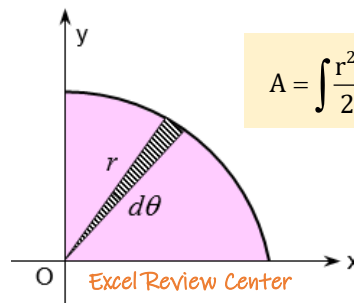
$$A = S \cdot 2\pi \bar{d}$$

where:

 S = length of arc d = distance from centroid to the axis of rotation

$$A = \int \sqrt{1 + \left(\frac{dy}{dx} \right)^2} \, dx \cdot 2\pi \bar{d}$$

B. Polar Coordinates

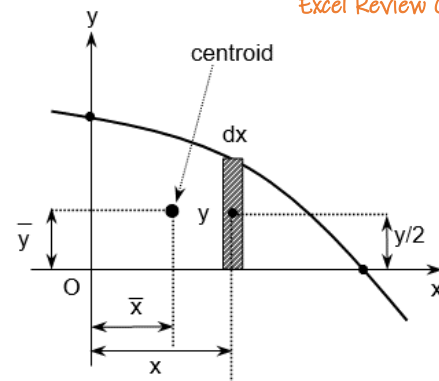


$$A = \int \frac{r^2}{2} \, d\theta$$

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2. Location of Centroid

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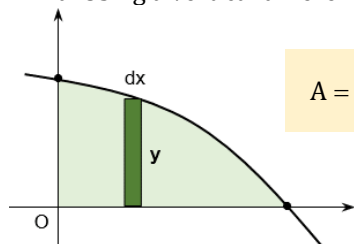
$$\bar{x} = \frac{\int dA \cdot x}{A}$$

$$\bar{y} = \frac{\int dA \cdot \frac{y}{2}}{A}$$

3. Length of Arc**Integral Calculus Applications****1. Area**

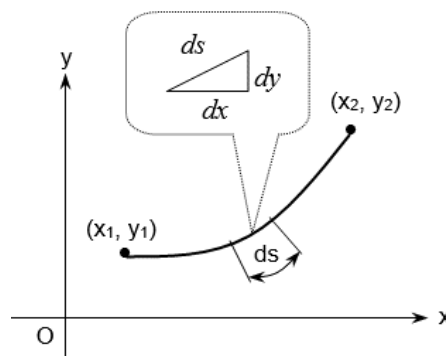
A. Rectangular Coordinates:

a. Using a vertical differential strip:



$$A = \int y \, dx$$

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5. Volume

The volume of a solid generated by revolving a curve about a certain axis may be calculated using the second proposition of Pappus. The **Second Proposition of Pappus** states as follows:

"If an area is rotated about an axis, it will generate a volume equal to the product of the area and the circumference described its centroid."

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